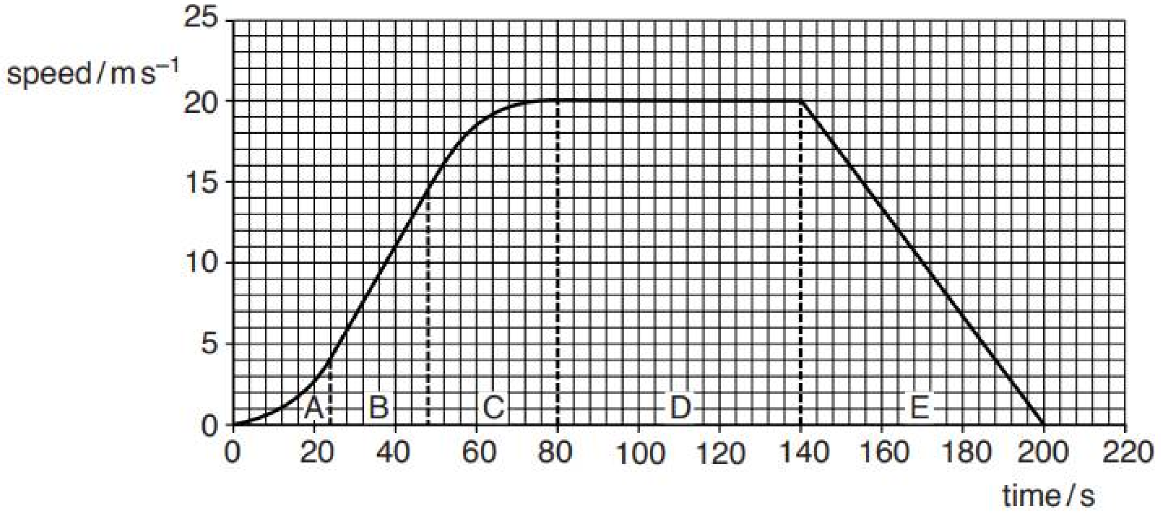
**1** **(a)** **(i)**




Distance = area of trapezium
= 20 (200 + 64) / 2
= 2640 m (accept 2580 m to 2700 m)

Accept other appropriate methods of estimation eg. counting squares.

**(ii)** Accept uncertainty 10 m to 130 m

Reason for uncertainty: (any one)

- Estimating area using straight lines instead of curves
- Difficulty of counting squares

**(b)** **(i)** $s_x = u_x t = (63 \cos 14)\ 4.9 = 299.5$ m
$s_y = u_y t + \frac{1}{2} a t^2 = (63 \sin 14)\ 4.9 + 0.5\ (-9.81)\ 4.9^2 = -43.09$ m
$\theta = \tan^{-1} (s_y / s_x) = 8.2°$ (to 2 sf)

**(ii)** **1.**


Features (all 3)
- At least 3 mm along original path **(but not too many mm)**
- New path under original path
- Max height is lower and earlier.

**(ii)** **2.** the path of the ball at the start, (any one)
- Path determined by movement of club / Same initial velocity
- Caused by same force in same direction
- Air resistance has acted for short time

the position for the highest point, (any one)
- Air resistance reduces upward velocity (and forward velocity)
- Air resistance opposes motion

the angle at which the ball hits the ground.
- Forward / horizontal velocity is (much) reduced

**2** **(a)** Current passing A = 0.50 / 5.0 = 0.10 A